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Dr. Héctor Carrión

World Model Team Lead · Multimodal & Spatio-Temporal AI

Function · Ph.D. in Computer Science, UC Santa Cruz / UC Berkeley

AI research lead (6+ YoE) building **multimodal world models** across longitudinal health, 3D vision, and autonomy. Leads the World Model team at **Function** (JEPA and autoregressive models over longitudinal labs and imaging); previously Founding Lead Research Scientist at **Tera AI** (1B-parameter road world model) and Ph.D. researcher at **Google Research**, **Google X** (granted patents), and **UC Santa Cruz / UC Berkeley**. First-author papers at **MICCAI** (×4) and **PLOS Comp. Biol.**

Focus: World Models • JEPA • 3D Vision • Diffusion • Medical AI • Embodied Autonomy

Research & Professional Experience

Sep 2026 – now
World Model Team Lead, Function
- Directing the World Model team and research roadmap on **JEPA** and **autoregressive** multimodal architectures over longitudinal lab tests, imaging, and member health history.
- Owning end-to-end pre-training and trajectory modeling to monitor early indicators and surface actionable member insights.

2024 – 2026
Founding Lead Research Scientist, Tera AI
- Set and executed the founding research agenda across spatial reasoning, generative models, and embodied autonomy, managing multi-node compute budgets and iteration schedules.
- Led **TARDIS** and **STRIDE** (with P. Perona and J. Malik): a 1B-parameter autoregressive spatio-temporal transformer unifying street-view synthesis, action prediction, and sequential georeferencing (60% within 10 m).

2020 – 2026
Graduate Student Researcher, UC Santa Cruz & UC Berkeley
- Built label-efficient medical vision models (<100 labeled samples) using controllable diffusion (**cgDDI**: 90.9% DDI accuracy, 86.6% EOM; **FEDD**: +18 IoU points) and monocular metric 3D reconstruction (**DermDepth**: 16.1× →1.11× SKINL2 depth scale ratio tuning 0.6% of parameters).
- Developed self-supervised temporal stage discovery (**HealNet**) and automated biomarker imaging pipelines for multi-institutional DARPA/NIH bioelectronic studies.

Summer 2023
Student Researcher (Ph.D.), Google Research
- Trained unified multimodal video-and-image foundation encoders, improving downstream transfer by >25% over deployed single-modality baselines.

2021 – 2022
AI Resident (Ph.D.), X, The Moonshot Factory (Google X)
- Built simulation, edge-perception, and aerial/ground computer vision systems for autonomous platforms, exceeding deployment benchmarks and earning granted **Google patents**.

Education

2020 – 2026
Ph.D. in Computer Science, University of California, Santa Cruz
Advisor: Dr. Narges Norouzi (UC Berkeley). GPA: 3.92 / 4.00.

2017 – 2021
B.Sc. in Computer Science, University of Puerto Rico, Río Piedras
Undergraduate Research PI: Dr. Rémi Mégret. GPA: 3.75 / 4.00.

2018 – 2019
Software Engineering & Design, Apple Developer Academy, Univ. of Naples Federico II
Full-stack iOS engineering, interface design, and on-device CoreML applications.

Publications

MICCAI 2026
DermDepth: Toward Monocular Metric Scale 3D Reconstruction Models for Dermatology
H. Carrión, N. Norouzi
16.1× →1.11× SKINL2 depth scale with 0.6% tuned params paper | arXiv | code | data | model | demo

MICCAI 2026
cgDDI: Controllable Generation of Diverse Dermatological Imagery for Fair and Efficient Malignancy Classification
H. Carrión, N. Norouzi
266K+ synthetic images; 90.9% DDI accuracy & 86.6% EOM paper | arXiv | code | data

Comm. Biol. 2025

A Modular Fluorescent Camera Unit for Wound Imaging

M. Tebyani, G. Keller, W. S. Hee, P. Baniya, …, **H. Carrión**, …, R. R. Isseroff, M. Rolandi, M. Teodorescu

Nature Communications Biology paper

arXiv 2025

TARDIS STRIDE: A Spatio-Temporal Road Image Dataset and World Model for Autonomy

H. Carrión, Y. Bai, V. A. Hernández Castro, K. Panaganti, A. Zenith, M. Trang, T. Zhang, P. Perona, J. Malik

1B-param world model; 60% within 10 m georeferencing arXiv | project | code | data

Adv. Mater. Tech.

Remote-Controlled Wireless Bioelectronics for Fluoxetine Therapy to Promote Wound Healing in a Porcine Model

H. Li, N. Asefifeyzabadi, …, **H. Carrión**, N. Norouzi, M. Gomez, M. Zhao, R. R. Isseroff, M. Teodorescu, M. Rolandi

Advanced Materials Technologies, 2025 paper

Adv. Mater. Tech.

Programmable Delivery of Fluoxetine via Wearable Bioelectronics for Wound Healing In Vivo

H. Li, H.-Y. Yang, N. Asefifeyzabadi, …, **H. Carrión**, …, N. Norouzi, M. Gomez, M. Zhao, M. Teodorescu, R. R. Isseroff, M. Rolandi

Advanced Materials Technologies, 2024 paper

MICCAI 2023

FEDD: Fair, Efficient, and Diverse Diffusion-Based Lesion Segmentation and Malignancy Classification

H. Carrión, N. Norouzi

81% DDI malignancy accuracy from 60 labeled images paper | arXiv | code

MICCAI-W 2022

HealNet: Self-Supervised Acute Wound Heal-Stage Classification

H. Carrión, M. Jafari, H.-Y. Yang, R. R. Isseroff, M. Rolandi, M. Gomez, N. Norouzi

Machine Learning in Medical Imaging (MLMI); 90.6% stage accuracy paper | arXiv

PLOS CB 2022

Automatic Wound Detection and Size Estimation Using Deep Learning Algorithms

H. Carrión, M. Jafari, M. D. Bagood, H.-Y. Yang, R. R. Isseroff, M. Gomez

PLOS Computational Biology paper | code

VISAPP 2022

Honeybee Re-Identification in Video: New Datasets and Impact of Self-Supervision

J. Chan, **H. Carrión**, R. Mégret, J. Rivera, T. Giray

109K-image self-supervised track dataset paper

Open-Source Code, Models & Datasets

DermDepth	Metric 3D weights, D-Synth dataset, and live Space demo	github.com/hectorcarrion/dermdepth
cgDDI	Hybrid diffusion code and 266K+ synthetic dermatology dataset	github.com/hectorcarrion/ControllableGenDDI
TARDIS	1B-param world model (JAX/EasyLM) and STRIDE dataset	github.com/tera-ai/tardis
FEDD	Diffusion feature probes and DDI lesion segmentation masks	github.com/hectorcarrion/fedd

Honors, Patents & Advisory

Patents	**Google Patent Award**: Granted perception & autonomous systems patents at X (Google X)
Fellowships	**Google–CAHSI Doctoral Fellowship** | **NIH BD2K Biomedical Data Science Fellowship**
Awards	**MICCAI STAR Award** | **SIDIM Outstanding Research Award**
Advisory	**Technical Advisor**, Visia AI (2021–present): computer vision for recyclability & hazard monitoring
Mentorship	**Machine Learning Tutor**, UC Santa Cruz (2020–2022): deep learning & computer vision

Skills & Languages

Programming	Python, C++, Swift, JAX, Bash, LaTeX
Deep learning	PyTorch, JAX/EasyLM, TensorFlow, Keras, JEPA & autoregressive world models, diffusion, multimodal LLMs
Domains	Longitudinal health modeling, metric 3D vision, medical imaging, spatial autonomy, fairness
Leadership	Research roadmap & team management, compute budgeting, cross-functional execution, product design
Languages	English (native), Spanish (native)

Glyph key: World Model · 3D Vision · Diffusion · Med AI · Self-Supervised / JEPA · Autonomy · Bioelectronics